STM32 Microcontroller Architecture & Electrical Interface

1. Overview of STM32F103C8T6 ("Blue Pill")

The STM32F103C8T6 is an entry-level high-performance 32-bit microcontroller manufactured by STMicroelectronics. It features the **ARM Cortex-M3 core (ARMv7-M architecture)** utilizing a 3-stage instruction pipeline (Fetch, Decode, Execute) with Harvard architecture (separate instruction and data buses: ICode and DCode).

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2. Electrical Specifications & Memory Footprint

Parameter	Datasheet Rating	Project Usage & Safety Limits
CPU Core	32-bit ARM Cortex-M3 (RISC)	72 MHz SYSCLK operating frequency
Flash Memory	64 KB (128 KB factory unlocked)	Binary size: ~29 KB (leaves >50% headroom)
SRAM	20 KB (20,480 bytes)	Uses ~4.6 KB (1 KB display buffer + variables)
Supply Voltage (V_{DD})	2.0 V to 3.6 V	Powered at 3.3V via onboard RT9193 LDO regulator
Max GPIO Current Output	25 mA per pin	LEDs limited to ~8 mA with $330\,\Omega$ series resistors
Total MCU Current Limit	150 mA sum of all I/Os	Entire system draws ~45 mA at full OLED brightness
5V Tolerant (FT) Pins	Up to 5.5V without damage	**PB6**, **PB7**, **PB12**, **PB14**, **PB15** are FT rated
Non-5V Tolerant Pins	Max $V_{DD} + 0.3\text{V} \approx 3.6\text{V}$	**PA0**-**PA7** (ADC channels), **PB0**, **PB1**

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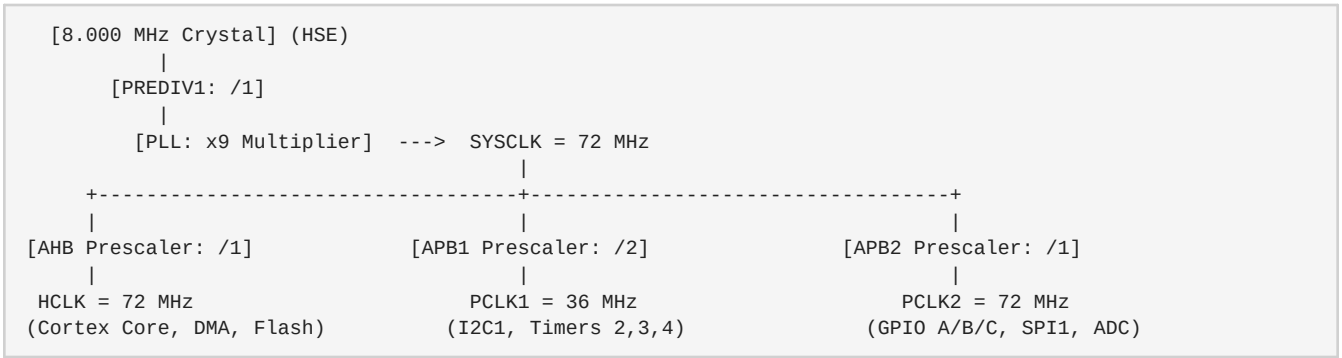
3. Clock Distribution Network & Phase-Locked Loop (PLL)

The STM32F103 clock tree configuration determines CPU execution speed and peripheral communication rates:



3.1 Mathematical Clock Derivations

1. **System Clock (f_{SYSCLK})**:

$$f_{\text{SYSCLK}} = f_{\text{HSE}} \times 9 = 8.000\,\text{MHz} \times 9 = 72.000\,\text{MHz}$$

2. **APB1 Peripheral Clock (f_{PCLK1})**:

$$f_{\text{PCLK1}} = \frac{f_{\text{HCLK}}}{2} = \frac{72.000\,\text{MHz}}{2} = 36.000\,\text{MHz}\quad (\text{Max allowed for APB1 is } 36\,\text{MHz})$$

3. **Flash Memory Latency (Wait States)**:

Flash access time is limited to $\le 24\,\text{MHz}$. At 72 MHz:

$$\text{Required Wait States} = \left\lceil \frac{72\,\text{MHz}}{24\,\text{MHz}} \right\rceil - 1 = 3 - 1 = 2\text{ Wait States (FLASH_LATENCY_2)}$$

Failing to configure 2 wait states causes instruction prefetch memory corruption and immediate **`HardFault_Handler`** execution.

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4. GPIO Electrical Architecture: Push-Pull vs. Open-Drain

Inside each STM32 GPIO pin cell, two complementary power MOSFETs control pin state:

```
          VDD (3.3V or 5V FT)
             |
          [P-MOS]  <--- Driven by Output Control Logic
             |
Pin ----+---+
        |
     [N-MOS]      <--- Driven by Output Control Logic
        |
       GND
```

4.1 Push-Pull Output Mode (**`GPIO_MODE_OUTPUT_PP`**)

- **Operation**: P-MOS conducts for Logic HIGH (3.3V); N-MOS conducts for Logic LOW (0V).
- **Used for**: **`PB1`** (Pill 1 LED), **`PB10`** (Pill 2 LED), **`PB0`** (Buzzer), **`PA4`** (SD CS), **`PA5`** (SCK), and **`PA7`** (MOSI).
- **Advantage**: Fast rise and fall times with symmetric drive capability ($25\,\text{mA}$).

4.2 Open-Drain Output Mode (**`GPIO_MODE_AF_OD`**)

- **Operation**: P-MOS is permanently deactivated. Pin can only pull to GND (N-MOS active) or enter High-Impedance (Hi-Z) when N-MOS is off. An external pull-up resistor pulls the bus to HIGH.
- **Used for**: **`PB6`** (I2C1 SCL) and **`PB7`** (I2C1 SDA).
- **Why it is Mandatory for I2C**: Enables **wired-AND** logic. If slave and master both transmit simultaneously, no high-current contention occurs. Slaves can also hold SCL LOW to request clock stretching.

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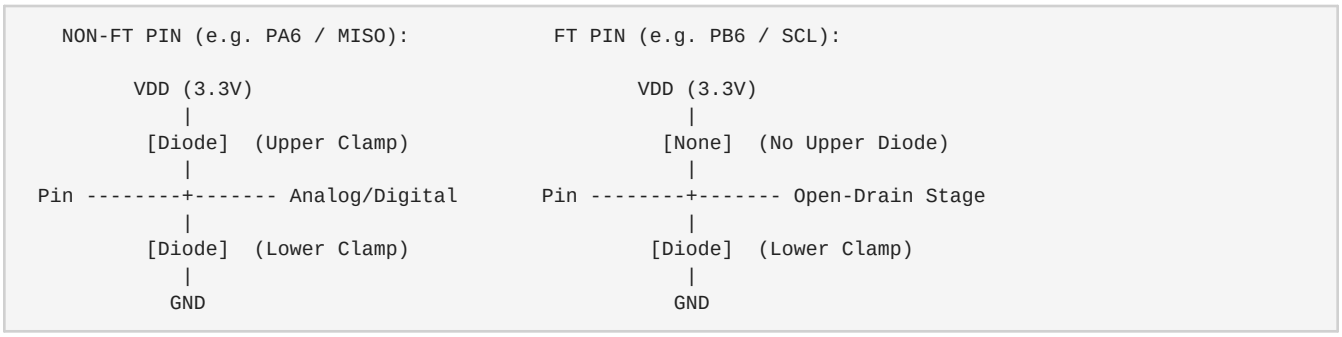
5. Five-Volt Tolerance (FT) vs. Non-FT Silicon Physics



• **Non-FT Pins (PA0-PA7, PB0, PB1)**: Incorporate an upper internal protection clamping diode tied to V_{DD} (3.3V). Applying 5V forward-biases this diode, causing high parasitic current into the 3.3V supply rail, latch-up, and thermal destruction of the analog multiplexer.

• **FT Pins (PB6, PB7, PB12, PB14, PB15)**: Have no upper diode connected to V_{DD}. Instead, they use a high-voltage N-MOS with a deep N-well breakdown rating exceeding 6.0V, allowing safe direct connection to 5V pull-ups when configured in open-drain mode.

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6. Evaluator Oral Questions & Detailed Answers

Q1: What is the RC rise-time equation for the I2C bus pull-up resistors?

***Answer**:*

In I2C open-drain mode, the rising edge is passive and governed by an RC charging curve:

$$V(t) = V_{DD} \left(1 - e^{-t / (R_p C_b)}\right)$$

The standard I2C specification defines rise time t_r from $0.3\,V_{DD}$ to $0.7\,V_{DD}$:

$$t_r \approx 0.8473 \times R_p \times C_b$$

For Standard Mode ($100\,\text{kHz}$), the maximum permitted rise time is $t_{r,\max} = 1000\,\text{ns}$. With a bus capacitance $C_b \approx 50\,\text{pF}$ (PCB trace + OLED + RTC pins), a $4.7\,\text{k}\Omega$ pull-up yields:

$$t_r = 0.8473 \times 4700 \times 50 \times 10^{-12} \approx 199\,\text{ns} \ll 1000\,\text{ns}$$

This provides crisp square waves with zero bit distortion.

Q2: Why did you not use interrupts (EXTI) for button detection?

***Answer**:*

*Mechanical pushbuttons exhibit contact bounce lasting 5 ms to 25 ms. If mapped directly to External Interrupts (**EXTI12**, **EXTI14**, **EXTI15**), a single press generates dozens of rapid nested ISR triggers, starving the main loop and causing stack overflow. Instead, we use a **10 ms periodic polling with software timestamp edge detection**:*

```c
if (lastState == HIGH && currentState == LOW) {
if (now - lastPressTime > 200) { /* Process valid click */ }
}
```

This is completely deterministic, consumes zero interrupt overhead, and eliminates bounce completely.

Q3: How is the SysTick hardware timer configured in this project?

> ***Answer**:*
>
> *SysTick is a 24-bit downcounter clocked by $HCLK = 72\,\text{MHz}$. The reload register is programmed to:*
>
> *$$\text{LOAD} = \frac{72\,\text{MHz}}{1000\,\text{Hz}} - 1 = 71{,}999$$*
>
> *When the counter reaches 0, it asserts the* **`SysTick_Handler`** *vector every $1.000\,\text{ms}$, incrementing* **`uwTick`**. *This gives* **`HAL_GetTick()`** *a millisecond timebase for all non-blocking timers.*